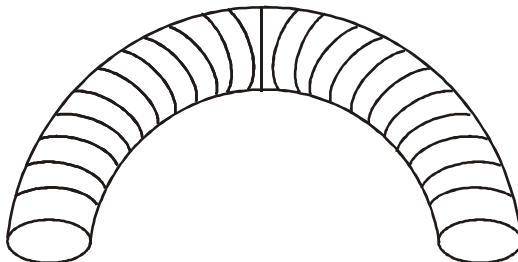




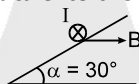
High Level Problems (HLP)

SUBJECTIVE QUESTIONS

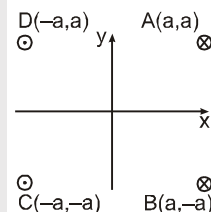
1. Calculate the magnetic moment of a thin wire with a current $I = 0.8$ A, wound tightly on half a toroid (figure). The diameter of the cross-section of the tore is equal to $d = 5.0$ cm, the number of turns is $N = 500$.



2. A small charged ball suspended on an inextensible thread of length L moves in a uniform time-independent vertically upward magnetic field of induction B . The mass of the ball is m and the charge is q , and angular speed is ω . Determine the radius of the horizontal circle in which the ball moves if the thread always remains taut. Take gravitational acceleration to be equal to g .
3. The figure shows a conductor of weight 1.0 N & length $L = 0.5$ m placed on a rough inclined plane making an angle 30° with the horizontal so that conductor is perpendicular to a uniform horizontal magnetic field of induction $B = 0.10$ T. The coefficient of static friction between the conductor and the plane is 0.1 . A current of $I = 10$ A flows through the conductor inside the plane of this paper as shown. What is the force needed to be applied parallel to the inclined plane to keep the conductor at rest?



4. 4 long wires each carrying current I as shown in the figure are placed at the points A, B, C and D. Find the magnitude and direction of:
(a) magnetic field at the centre of the square
(b) force per metre acting on wire at point D.

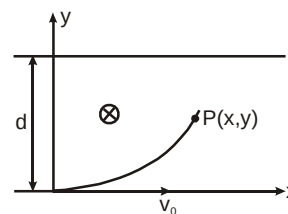


5. A uniformly charged ring of radius 0.1 m rotates at a frequency of 10^4 rps about its axis. Find the ratio of energy density of electric field to the energy density of the magnetic field at a point on the axis at distance 0.2 m from the centre. (Use speed of light $c = 3 \times 10^8$ m/s, $\pi^2 = 10$)
6. The current density $\vec{J}$ inside a long, solid, cylindrical wire of radius $a = 12$ mm is in the direction of the central axis, and its magnitude varies linearly with radial distance r from the axis according to $J = \frac{J_0 r}{a}$, where $J_0 = \frac{10^5}{4\pi}$ A/m². Find the magnitude of the magnetic field at $r = \frac{a}{2}$ in μ T.
7. A neutral particle is at rest in a uniform magnetic field $\vec{B}$. At $t = 0$, particle decays into two particles each of mass ' m ' and one of them having charge ' q '. Both of these move off in separate paths lying in plane perpendicular to $\vec{B}$. At later time, the particles collide. Find this time of collision neglecting the interaction force.



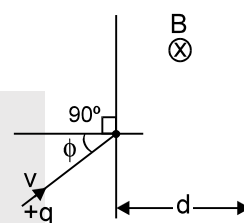


8. A non-uniform magnetic field $\vec{B} = B_0 \left(1 + \frac{y}{d}\right) (-\hat{k})$ is present in region of space in between $y = 0$ & $y = d$. The lines are shown in the diagram. A particle of mass 'm' and positive charge 'q' is moving. Given an initial velocity $\vec{V} = v_0 \hat{i}$. Find the components of velocity of the particle when it leaves the field.

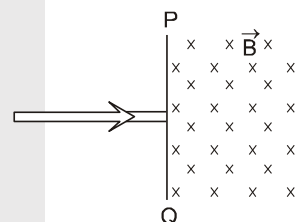


9. A neutral atom of atomic mass number 100 which is stationary at the origin in gravity free space emits an α -particle (A) in z-direction. The product ion is P. A uniform magnetic field exists in the x-direction. Disregard the electro magnetic interaction between A and P. If the angle of rotation of A after which A and P will meet for the first time is $\frac{n\pi}{25}$ radians, what is the value of n?

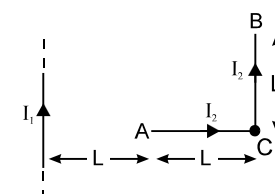
10. In the figure shown a positively charged particle of charge 'q' and mass 'm' enters into a uniform magnetic field of strength 'B' as shown in the figure. The magnetic field points inwards and is present only within a region of width 'd'. The initial velocity of the particle is perpendicular to the magnetic field and $\phi = 30^\circ$. Find the time spent by the particle inside the magnetic field if $d = 0.2\text{m}$, $B = 1\text{T}$, $q = 1\text{C}$, $m = 1\text{kg}$ and $v = 1\text{ m/s}$. Use $\sin 45^\circ = 0.7$ if required.



11. A thin beam of charged particles is incident normally on the boundary of a region containing a uniform magnetic field as shown. The beam comprise of mono energetic α and β^- particles, each possessing a kinetic energy T. Neglecting interaction between particles of the beam, find the separation between the points on line PQ where the α and β particles emerge out of the magnetic field. (express your answer in terms of T, B, α ($2e, m_\alpha$) & β^- ($-e, m_e$), where all terms have their usual meanings.)

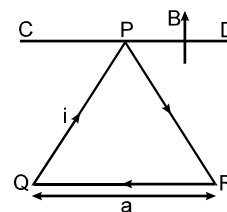


12. An electron moving with velocity $\vec{v}_1 = 1\hat{i} \text{ m/s}$ at a point in a magnetic field experiences a force $\vec{F}_1 = -e\hat{j} \text{ N}$, where e is the charge of electron. If the electron is moving with a velocity $\vec{v}_2 = \hat{i} - \hat{j} \text{ m/s}$ at the same point, it experiences a force $\vec{F}_2 = -e(\hat{i} + \hat{j}) \text{ N}$. Find the magnetic field and the force the electron would experience if it were moving with a velocity $\vec{v}_3 = \vec{v}_1 \times \vec{v}_2$ at the same point.
13. An electron is shot into one end of a solenoid. As it enters the uniform magnetic field within the solenoid, its speed is 800 m/s and its velocity vector makes an angle of 30° with the central axis of the solenoid. The solenoid carries 4.0 A current and has 8000 turn along its length. Find number of revolutions made by the electron within the solenoid by the time it emerges from the solenoid's opposite end. (Use charge to mass ratio e/m for electron $= \sqrt{3} \times 10^{11} \text{ C/kg}$) Fill your answer in multiple of 10^3 . (neglect end effect)
14. In the figure shown an infinitely long wire carries a current I_1 and another uniform rigid wire ACB (bent at 'C' at right angle) of mass 'm' carries a current I_2 , it is hinged at corner 'C'. Find the angular acceleration of this wire ACB just after release. Is the direction of rotation clockwise or anticlockwise. (Assume gravity is absent) [Take $\ln 2 = 0.7$]





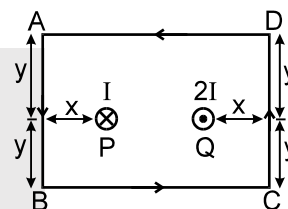
15. A loop PQR formed by three identical uniform conducting rods each of length 'a' is suspended from one of its vertices (P) so that it can rotate about horizontal fixed smooth axis CD. Initially plane of loop is in vertical plane. A constant current 'i' is flowing in the loop. Total mass of the loop is 'm'. At $t = 0$, a uniform magnetic field of strength B directed vertically upwards is switched on. Acceleration due to gravity is 'g'. Then find the minimum value of B so that the plane of the loop becomes horizontal (even for an instant) during its subsequent motion.



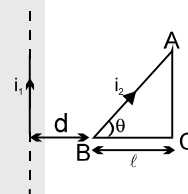
16. In order to impart an angular velocity to an earth satellite the geomagnetic field can be used. Find the maximum possible angular velocity about its own axis gained by the satellite if a storage battery with a capacity of $Q = 5$ Amp. hours is discharged suddenly through a coil of $N = 20$ turns wound around the satellite's surface along the circumference of the largest circle. The satellite has a mass of $m = 10^3$ kg and is a thin walled uniform sphere. The geomagnetic field is parallel to the winding plane and its flux density is $B = 0.5$ Gauss. ($1 \text{ Gauss} = 10^{-4} \text{ Tesla}$)

17. Let B_P and B_Q be the magnetic field produced by the wire P and Q which are placed symmetrically in a rectangular loop ABCD as shown in figure. Current in wire P is I directed inward and in Q is $2I$ directed outwards.

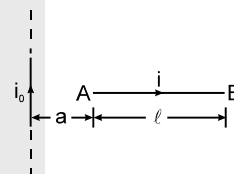
if $\int_A^B \vec{B}_Q \cdot d\vec{\ell} = 2\mu_0$ tesla meter, $\int_D^A \vec{B}_P \cdot d\vec{\ell} = -2\mu_0$ tesla meter & $\int_A^B \vec{B}_P \cdot d\vec{\ell} = -\mu_0$ tesla meter. Then find the value of I



18. A triangular loop (ABC) having current i_2 and an infinite wire having current i_1 are placed in the same plane. Find the magnetic force of interaction between the infinite wire and the loop ABC.

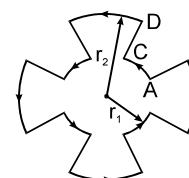


19. A finite conductor AB carrying current i is placed near a fixed very long wire current carrying i_0 as shown in the figure. Find the point of application and magnitude of the net ampere force on the conductor AB. What happens to the conductor AB if it is free to move? (Neglect gravitational field)



20. A particle of mass m and charge q is moving in a region where uniform, constant electric and magnetic $\vec{E}$ & $\vec{B}$ fields are present, $\vec{E}$ & $\vec{B}$ are parallel to each other. At time $t = 0$ the velocity of the particle is perpendicular to $\vec{E}$. (Assume that its speed is always $\ll c$, the speed of light in vacuum). Find the velocity $\vec{v}$ of the particle at time t. You must express your answer in terms of t, q, m, the vectors $\vec{v}_0$, $\vec{E}$ & $\vec{B}$ and their magnitudes v_0 , E and B.

21. A current of 10A flows around a closed path in a circuit which is in the horizontal plane. The circuit consists of eight alternating arcs of radii $r_1 = 0.08$ m and $r_2 = 0.12$ m. Each arc subtends the same angle at the centre.

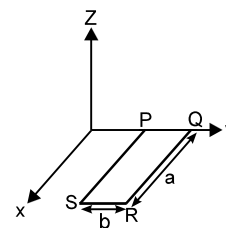


- Find the magnetic field produced by this circuit at the centre.
- An infinitely long straight wire carrying a current of 10 A is passing through the centre of the given circuit vertically with the direction of the current being into the plane of the circuit. What is the force acting on the wire at the centre due to the current in the circuit? What is the force acting on the arc AC and the straight segment CD due to the current in the central wire?

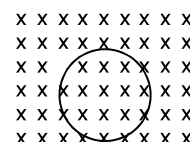




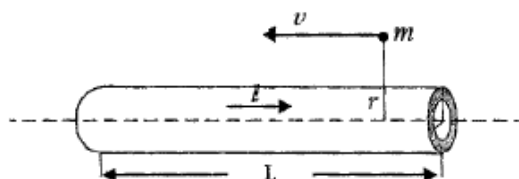
22. A rectangular loop PQRS made from a uniform wire has length a , width b and mass m . It is free to rotate about the arm PQ, which remains hinged along a horizontal line taken as the y -axis as shown in figure. Take the vertically upward direction as the z -axis. A uniform magnetic field $\vec{B} = (3\hat{i} + 4\hat{k})B_0$ exists in the region. The loop is held in the x - y plane and a current I is passed through it. The loop is now released and is found to stay in the horizontal position in equilibrium.
- (a) What is the direction of the current I in PQ?
- (b) Find the magnetic force on the arm RS.
- (c) Find the expression for I in terms of B_0 , a , b and m .



23. A ring of mass m and radius r is rotated in uniform magnetic field B which is perpendicular to the plane of the loop with constant angular velocity ω_0 . Find the net ampere force on the ring and the tension developed in the ring if there is a current i in the ring. Current and rotation both are clockwise.



24. A positive charge particle of charge ' q ' & mass ' m ' is released at origin. There are uniform magnetic field and electric field in the space given by $\vec{E} = E_0\hat{j}$ & $\vec{B} = B_0\hat{k}$, where E_0 & B_0 are constants. Find the ' y ' co-ordinate of the particle at time ' t '.
25. A square loop of wire of edge a carries a current i .
Magnetic induction B for a point on the axis of the loop at a distance $x = \frac{a}{\sqrt{2}}$ from its centre is $\frac{N\mu_0 i}{3\pi a}$, then find value of N .
26. A positive point charge q of mass m , kept at a distance x_0 (in the same plane) from a fixed very long straight current is projected normally away from it with speed v . Find the maximum separation between the wire and the particle.
27. Consider a solid sphere of radius r and mass m which has a charge q distributed uniformly over its volume. The sphere is rotated about a diameter with an angular speed ω . Find magnetic moment of the sphere.
28. A long non-magnetic cylindrical conductor with inner radius a and outer radius b carries a current I . The current density in the conductor is uniform. Assume end effects can be neglected. [Olympiad 2013; Stage-2]



- (a) Prove by the Biot-Savart law that the magnetic field, if any, at a point is always tangential to the circle passing through the point, with the centre of the circles on the axis of the cylinder.
- (b) Find the magnetic field due to the current as a function of radius.
- inside the hollow space ($r < a$)
 - within the conductor ($a < r < b$)
 - outside the conductor ($r > b$)
- (c) A beam of particles, each with positive charge q and mass m travels with initial velocity v anti-parallel to the direction of the current. Assume that the length of cylinder is L . Find the deflection of a particle as a function of its initial distance r from the axis if $r > b$ when it goes from one end to other end. You can assume that the velocity is high enough such that the velocity is constant and the deflection is small.





HLP Answers

1. $\frac{1}{2} \text{ A.m}^2$
2. $r = \left[L^2 - \left\{ \frac{mg}{m\omega^2 \pm q\omega B} \right\}^2 \right]^{1/2}$
3. $\frac{3}{4} \left(1 - \frac{\sqrt{3}}{10} \right) \text{ N to the left}$ and $\frac{3}{4} \left(1 + \frac{\sqrt{3}}{10} \right) \text{ N}$
4. (a) $\frac{\mu_0}{4\pi} \left(\frac{4I}{a} \right)$ along Y-axis (b) $\frac{\mu_0 I^2 (-\hat{j} - 3\hat{i})}{8\pi a}$
5. $9 \times 10^9 \text{ J}$
6. 10
7. $\frac{\pi m}{qB}$
8. $V_x = V_0 - \frac{3qB_0 d}{2m}$ and $V_y = \sqrt{V_0^2 - \left(V_0 - \frac{3qB_0 d}{2m} \right)^2}$
9. 48
10. $\frac{\pi}{12} \text{ sec.}$
11. $\frac{2\sqrt{2T}}{eB} \left\{ \frac{\sqrt{m_a}}{2} + \sqrt{m_e} \right\}$
12. $\vec{B} = -\hat{k}, \vec{F} = 0$
13. 1600
14. $\frac{9\mu_0 I_1 I_2}{40\pi mL}$ a.c.w.
15. $\frac{4mg}{3ia}$
16. $\omega = \frac{3BN\pi Q}{2M} = 2.7\pi \times 10^{-2} \text{ rad/s.}$
17. 6A
18. $\frac{\mu_0 i_1 i_2 \tan \theta}{2\pi} \left[\ln \frac{d+\ell}{d} - \frac{\ell}{d+\ell} \right]$ towards left.
19. $F = \frac{\mu_0}{2\pi} (i_0 i) \ln \left(1 + \frac{\ell}{a} \right)$ in the direction of i_0 . $x = \frac{\ell}{\ln \left(1 + \frac{\ell}{a} \right)}$, where x is the perpendicular distance from the wire i_0 . It will move upward and will rotate in the clockwise direction.
20. $\vec{v}(t) = \cos \omega t \vec{v}_0 + \frac{(\sin \omega t)(\vec{v}_0 \times \vec{B}_0)}{B} + \frac{qt}{m} \vec{E}$ where $\omega = qB/m$
21. (i) $B_C = \frac{5\pi \times 10^{-4}}{24} \text{ T}$ (ii) F on central wire = 0 ; $F_{AC} = 0$; $F_{CD} = 2 \times 10^{-5} \ln \left(\frac{3}{2} \right)$
22. (a) from P to Q (b) $lbB_0 (3\hat{k} - 4\hat{i})$ (c) $\frac{mg}{6bB_0}$
23. $0, \frac{r}{2\pi} (m\omega_0^2 + 2\pi iB)$
24. $y = \frac{E_0 m}{qB_0^2} \left[1 - \cos \frac{qB_0}{m} t \right]$
25. N = 2
27. $\frac{1}{5} q\omega R^2$